

Forecast-Aware Rolling-Horizon Optimization for Hybrid Wind-Storage Dispatch

David Valenta, North Carolina State University | Dr. Chris Qin, Washington State University
Summer 2026 REU | Pyron wind farm case study | ERCOT LMP data | PNNL CAES assumptions

Main takeaway
A forecast-aware controller must tune forecast horizon, replanning frequency, storage constraints, and scenario count together.

1. Problem

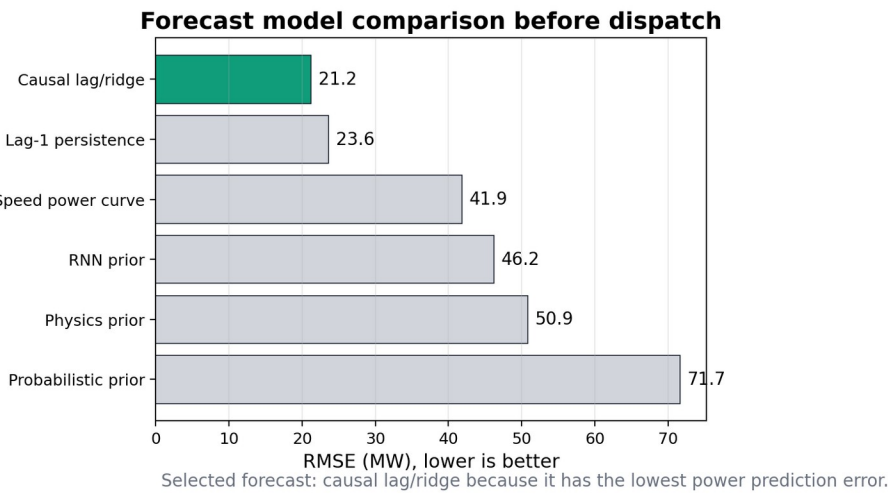
- Wind and electricity price change every hour.
- A hybrid wind farm can sell wind now or store energy for later.
- The challenge is dispatching storage before the future is known.
- The main comparison is the 100 MW constant-output CAES benchmark.
- Wind-only is kept as secondary reference information.

Why storage helps
If wind arrives during a low-price hour, storage can hold part of that energy and release it when the price is higher. The dispatch problem is deciding when that trade is worth it while keeping the battery feasible.

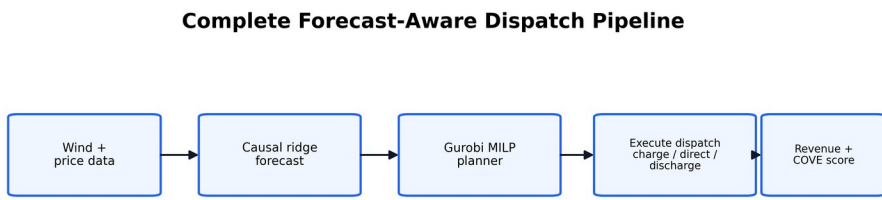
2. Data and Storage

- Pyron wind farm hourly data, 1980-2023.
- Forecast trained before 2014; tested on unseen 2014-2023.
- ERCOT locational marginal price is used for value.
- CAES: 100 MW, 10 h, 1000 MWh.
- SoC: 200-1000 MWh, starting at 600 MWh.
- RTE: 55%, no grid charging, 249 MW grid cap.

What the model sees
The controller does not receive the true future in deployable cases. It receives forecasted wind generation and forecasted price, then the realized data is used later for scoring.



Forecast result: causal lag/ridge was the best power forecast, with 21.24 MW RMSE.

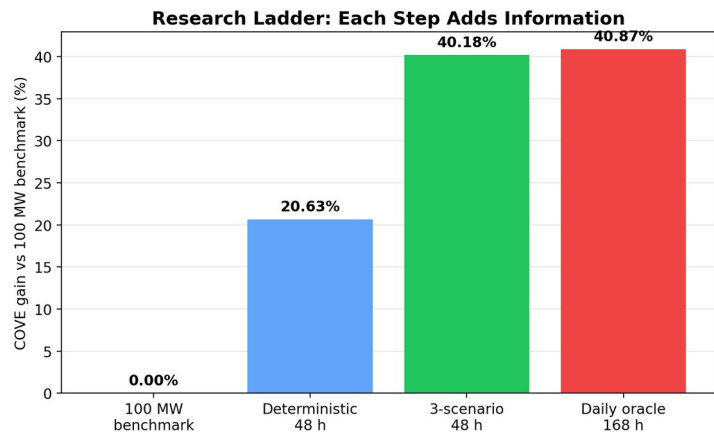


The pipeline predicts wind/price, sends those forecasts to Gurobi, executes dispatch, then scores revenue and COVE.
The important idea is separation: forecasts are guesses, Gurobi is the planner, and realized operation is the final truth used for revenue and COVE.

3. MILP Dispatch

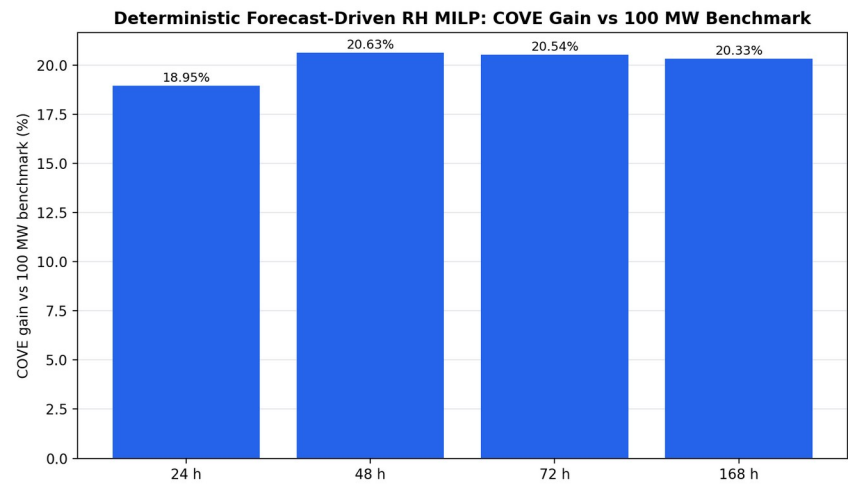
- Gurobi chooses direct wind, charge, discharge, SoC, and curtailment.
- Charging must come from wind, not the grid.
- The battery cannot charge and discharge beyond its physical limits.
- Revenue is delivered power times realized price.
- COVE is lower when energy is delivered at higher value.

Constraint	Value
Storage power	100 MW charge / 100 MW discharge
Capacity	1,000 MWh
SoC bounds	200-1,000 MWh
Initial SoC	600 MWh
RTE	55%, discharge side
Grid cap	249 MW
Grid charging	Not allowed

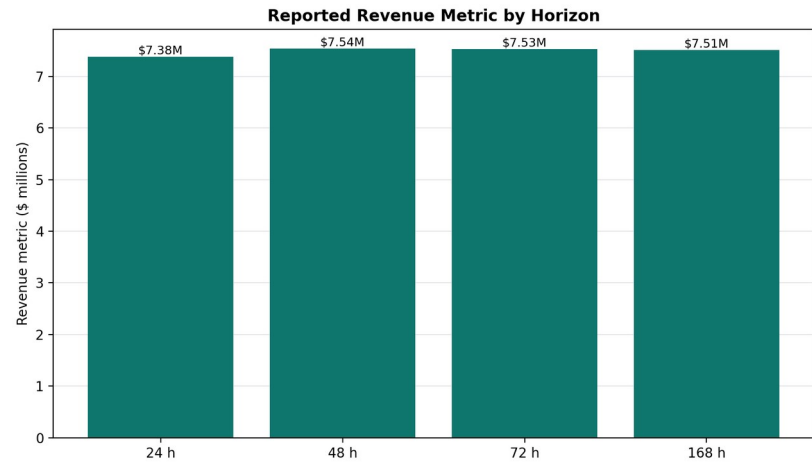


The ladder is not just decoration: each step adds one layer of information or control. Forecasting answers what may happen; rolling horizon decides how far to plan; scenarios add uncertainty; oracle shows the ceiling.

Part	What changed	Best output
Forecast	Predict wind power from past values	21.24 MW RMSE
Rolling horizon	One forecast path into Gurobi	48 h: 20.63% COVE gain
Scenarios	Multiple possible futures	3 scenarios: 40.18% COVE gain
Oracle	Perfect future information	168 h: 40.87% COVE gain



Deterministic rolling horizon: 48 hours gave the best COVE gain because it looked far enough ahead without trusting a weak long forecast too much.



Longer horizons did not automatically improve realistic dispatch. Planning value and forecast error have to balance.

How to read these results

A higher COVE gain is better because it means the same storage system delivered energy at higher value compared with the 100 MW benchmark.

4. Results

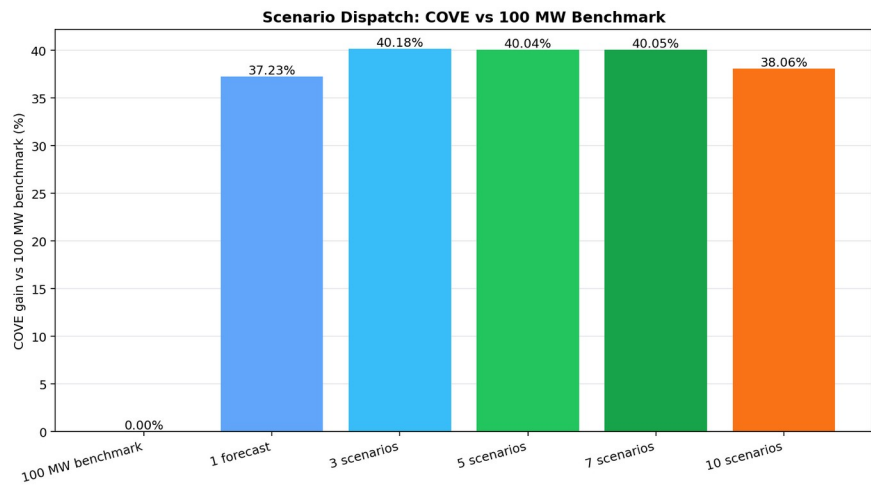
Primary benchmark: 100 MW constant-output CAES. Wind-only remains secondary context.

21.24 MW
Forecast RMSE
causal lag/ridge

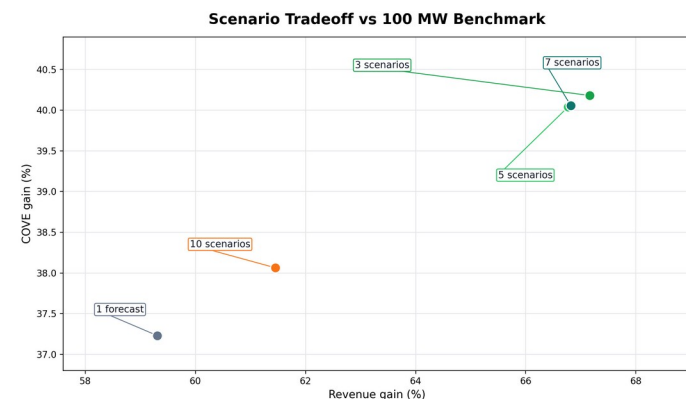
20.63%
Deterministic
48 h COVE gain

40.18%
Scenario
3 scenarios, hourly replan

40.87%
Oracle
168 h perfect future



Scenario result: 3 scenarios performed best. More scenarios became more cautious and did not add enough value.



Revenue and COVE move together here: the strongest scenario case both earns more and lowers COVE against the benchmark.

5. Main Conclusion

- The strongest realistic result came from the scenario-based controller because it planned under multiple possible futures and corrected decisions quickly.
- The deterministic 48 h case is useful because it shows the best one-forecast version of the method.
- The oracle is not realistic, but it shows the upper bound if future wind and price were known perfectly.
- The core claim is not just that Gurobi optimizes dispatch. The key result is that forecast design and dispatch design have to be tuned together.
- This matters for real wind farms because storage decisions are only useful when they respect uncertainty and physical constraints at the same time.

Best realistic method
3 scenarios with hourly replanning

Why it worked
It reacted faster to forecast mistakes.

Oracle meaning
It is the perfect-future ceiling.

Best result: 3-scenario forecast-aware rolling-horizon dispatch improved COVE by 40.18% vs the 100 MW benchmark.

Definitions and Acknowledgements

- 100 MW benchmark: rule-based storage case that tries to deliver 100 MW every hour.
- Rolling horizon: solve the future, execute the near-term action, update SoC, then solve again.
- Scenario dispatch: solve several possible futures instead of one forecast.
- Oracle: perfect future information, used only as a ceiling.
- COVE: cost divided by valued delivered energy; lower is better.
- All reported runs use chronological SoC and checked storage/grid constraints.

Thank you to Dr. Chris Qin, Nora Hosseiniemi, Zach Lawrence, Jessica Yao, and the WSU Summer Research Program. Data and assumptions draw from the Pyron wind-storage dataset, ERCOT LMP records, and PNNL CAES storage references.

What I tested
Forecast models, deterministic horizons, scenario counts, oracle ceiling, and constraint checks.

Contact: dvalent2@ncsu.edu
Poster size: 30 in x 20 in